Stability and Bifurcation of a Simple Food Chain in a Chemostat with Removal Rates

Mathematical Modelling Project Presentation

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Presentation Roadmap

1. Biological motivation
2. Mathematical model
3. Equilibria and stability
4. Numerical evidence
5. Persistence and coexistence
6. Hopf bifurcation
7. Effect of removal rates
8. Main conclusion

Core message

Distinct removal rates change the qualitative long-run behavior of the food chain.

Problem and Key Question

Biological setting

A chemostat continuously adds nutrient and removes culture medium, keeping volume approximately constant.

Food chain

$$S \longrightarrow x \longrightarrow y \longrightarrow z$$

- S: nutrient
- x: prey
- y, z: first and second predators

Main question

- Which species survive?
- When is coexistence stable?
- When do oscillations appear?
- How do D_1, D_2, D_3 reshape the outcome?

Mathematical Model

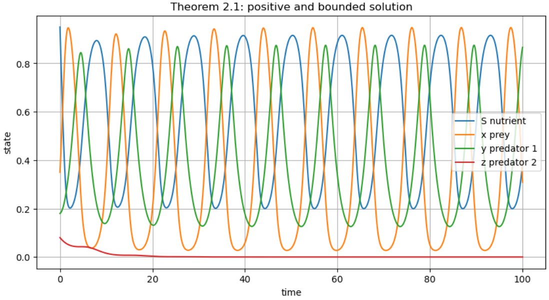
$$\frac{\mathrm{d}S}{\mathrm{d}t} = 1 - S - f_1(S)x, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = y\big(f_2(x) - D_2\big) - f_3(y)z,$$
$$\frac{\mathrm{d}x}{\mathrm{d}t} = x\big(f_1(S) - D_1\big) - f_2(x)y, \qquad \frac{\mathrm{d}z}{\mathrm{d}t} = z\big(f_3(y) - D_3\big).$$

- S, x, y, z: Nutrient, prey, 1st and 2nd predator
- f_i: Monod response functions (e.g., $a_i u/(b_i + u)$)
- D_1, D_2, D_3: Distinct removal rates
- **Key feature:** Distinct D_i removes simple conservation laws, requiring full 4D analysis.

Numerical Setup & Basic Properties

Workflow

- Solve ODEs using adaptive Runge-Kutta.
- Discard transient behavior.
- Classify long-run regimes (Washout, Coexistence, etc.)



S, x, y, z remain positive and bounded.

- **Positivity:** $\min(S, x, y, z) \approx 2.02 \times 10^{-9} > 0$.
- **Dissipativity:** $1.12 \leq S + x + y + z \leq 1.62$ (bounded, no divergence).

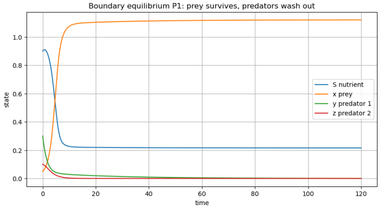
Equilibrium Points

Equilibrium	Form	Biological meaning
P_0	$(1, 0, 0, 0)$	Washout of all organisms
P_1	$(S_1, x_1, 0, 0)$	Prey survives; predators extinct
P_2	$(S_2, x_2, y_2, 0)$	Prey and first predator survive
P_3	(S_3, x_3, y_3, z_3)	Full coexistence

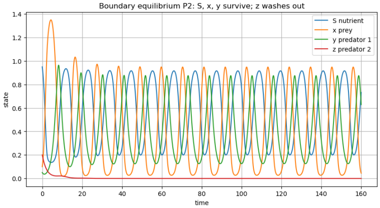
Interpretation

Equilibria are not only algebraic solutions; they represent possible ecological outcomes.

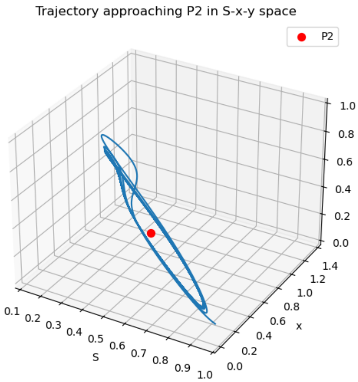
Boundary Regimes: Time-Series & Phase View



P_1: prey-only



P_2: top predator washed out



P_2 boundary phase portrait

- Different parameter conditions remove different trophic levels.
- Boundary equilibria correspond to real long-run regimes, showing attraction to lower-dimensional states.

Boundary Stability Checks

Eq.	Main condition	Value	Interpretation
P_0	Prey invasion at nutrient-only state	-0.1333	Washout stable
P_1	First-predator invasion at prey-only state	-0.0351	Prey-only survival
P_2	Top-predator invasion $f_3(y_2) - D_3$	-0.1178	z cannot persist

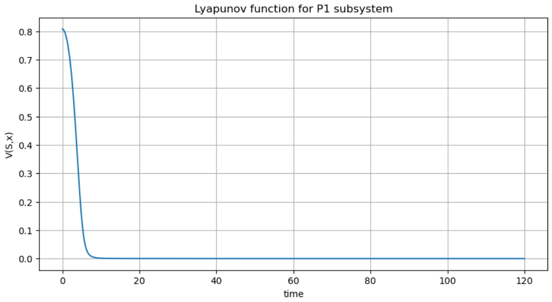
Careful interpretation

The P_2 value only says the top predator cannot invade near that boundary state.

Global Stability via Lyapunov Functions

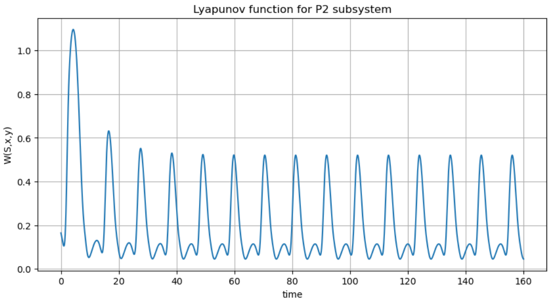
Prey-only subsystem

$$V(S,x) = \frac{1}{2}(S - S_1)^2 + \frac{1}{2}(x - x_1)^2$$



Boundary subsystem (P_2)

$$W = S - S_2 - S_2 \ln \frac{S}{S_2} + \frac{1}{2}(x - x_2)^2 + \frac{1}{2}(y - y_2)^2$$



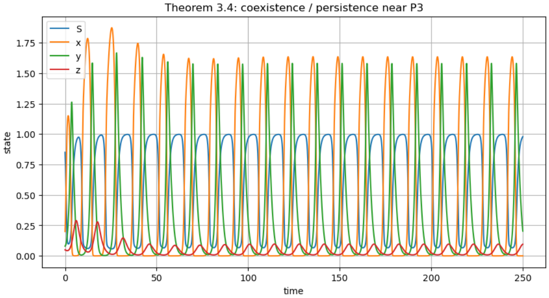
- Both profiles decrease toward small values, supporting global stability from wider regions.

Persistence and Coexistence at P_3

Coexistence equilibrium

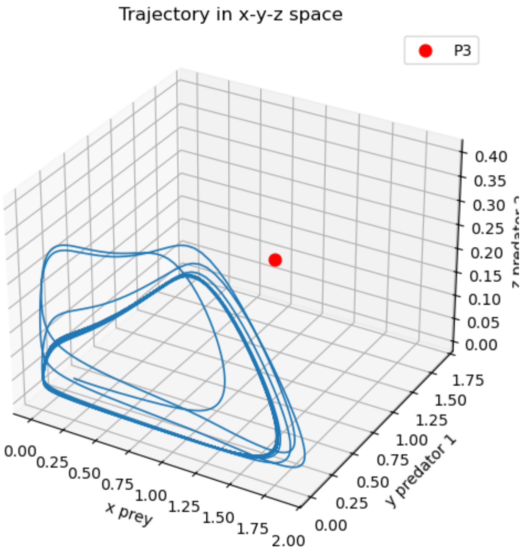
$$P_3 = (0.0669, 1.6291, 0.1667, 0.3963).$$

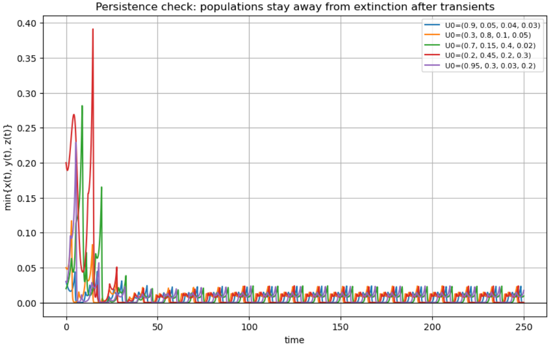
- All four components are positive.
- Persistence means populations stay away from extinction in the long run.



Coexistence time series

Coexistence: Phase and Robustness Checks





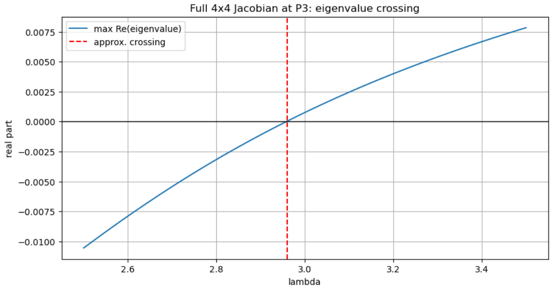
Multiple positive initial conditions

Hopf Bifurcation: Transition to Oscillations

Concept

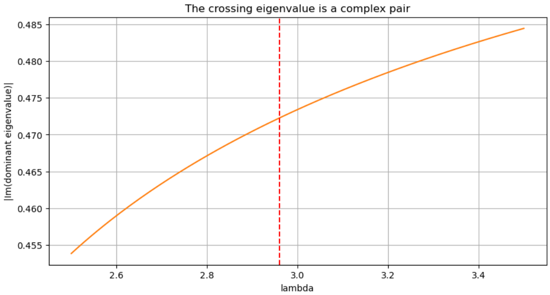
A Hopf bifurcation occurs when a complex conjugate pair of eigenvalues crosses the imaginary axis.

Before crossing (stable):
Equilibrium is an attractor. Populations settle to constants.



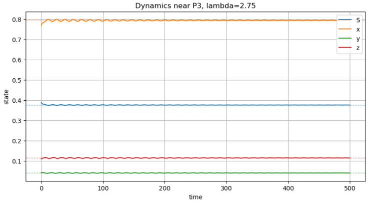
Real part crossing (near P_3)

After crossing (oscillatory):
Cyclic coexistence. Populations persist in sustained oscillations.

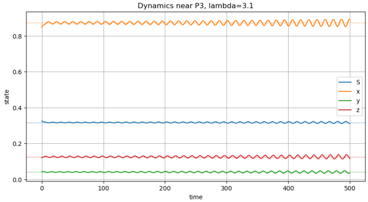


Complex conjugate pair

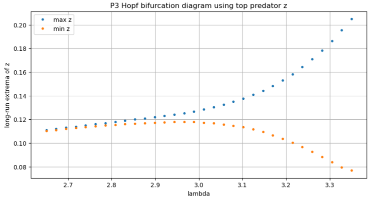
Hopf Bifurcation: Coexistence Dynamics (near P_3)



$\lambda = 2.75$: stable side



$\lambda = 3.10$: oscillatory side

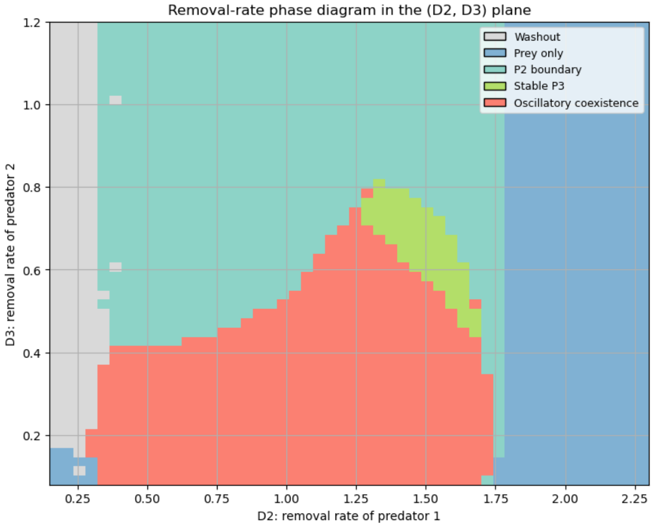


Top-predator extrema

Ecological interpretation

All species can persist, but their densities may oscillate instead of settling to constants depending on $a_1 = \lambda$.

Effect of Removal Rates: Phase Diagram

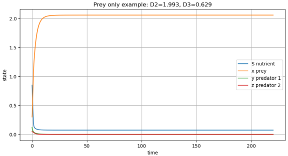


Parameter sweep

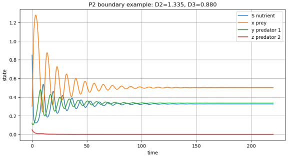
$$D_2 \in [0.15, 2.30], \qquad D_3 \in [0.08, 1.20].$$

- Grid size: 50×50.
- Five long-run regimes are classified.
- D_2, D_3 strongly shape survival and oscillation.

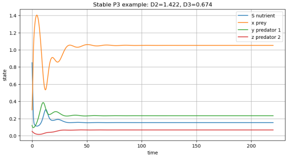
Representative Regimes from the Removal-Rate Sweep



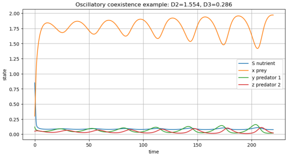
Prey-only



P_2 boundary



Stable P_3



Oscillatory coexistence

- Large predator removal rates eliminate higher trophic levels.
- Some regions support full coexistence; others produce cycles.

Conclusion and Limitations

Main conclusions

- Positivity and boundedness are numerically supported.
- Boundary regimes match invasion conditions.
- Hopf bifurcation appears near both P_2 and P_3.
- Distinct removal rates reshape long-run ecological outcomes.

Limitations

- No empirical calibration data.
- Deterministic model only.
- No delay, noise, space, or seasonality.
- Stability conditions may be sufficient, not necessary.

Distinct removal rates are mathematically and biologically meaningful.